IOE413 Proposal - Base Papers

Team-19

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Topic: Continual Learning for Concept-Drifting Data Streams

Track: B - Streaming ML for Anomaly & Drift

1. Problem Statement

Industrial IoT systems continuously generate high-velocity sensor streams where the underlying data distribution changes over time due to equipment aging, environmental variations, and operational shifts. These concept drifts degrade the performance of static machine learning models. The objective of this project is to build a real-time streaming analytics pipeline that detects concept drift, continually updates the learning model, and evaluates adaptation performance with low latency and high throughput.

2. Base Papers

Paper 1 (Baseline): ADWIN-U: adaptive windowing for unsupervised drift detection on data streams

Authors: Daniel Nowak Assis, Vinicius M. A. Souza (2025)

This paper proposes ADWIN-U, an unsupervised extension of the widely used ADWIN drift detector. Unlike traditional ADWIN, which requires labeled data to monitor model performance, ADWIN-U detects concept drift by monitoring statistical properties of the incoming stream such as skewness and kurtosis. The paper also introduces a new evaluation metric called BAR (Balanced Accuracy by Requested Labels) that measures the trade-off between drift detection quality and labeling cost, making it suitable for real-world IoT streams where labels are scarce.

Requirement	Details
Research Question	Can concept drift be detected accurately without labeled streaming data?
Data Used	10 real-world non-stationary data stream datasets
Metrics Used	Drift accuracy, False alarms, BAR, Requested labels
Key Findings	ADWIN-U outperformed supervised ADWIN while reducing dependence on labeled data. s10115-025-02523-1.pdf
Model	ADWIN-U

Paper 2: Gradient boosted bagging for evolving data stream regression

Authors: Nuwan Gunasekara, Bernhard Pfahringer, Heitor M. Gomes, Albert Bifet (2025)

This paper introduces Streaming Gradient Boosted Regression (SGBR), a continual learning ensemble method designed specifically for evolving data streams. The authors combine gradient boosting with streaming bagging to reduce prediction variance while maintaining adaptability under concept drift. Experiments across 11 streaming regression datasets demonstrate that SGBR consistently achieves higher predictive accuracy and lower computational cost than previous state-of-the-art streaming regression methods.

Requirement	Details
Research Question	Can gradient boosting improve continual learning for evolving data streams?
Data Used	11 streaming regression benchmark datasets
Metrics Used	R^2, Predictive accuracy, Runtime, Computational cost
Key Findings	SGBR achieved higher predictive power than previous state-of-the-art streaming regressors. s10618-025-01147-x.pdf
Model	Streaming Gradient Boosted Regression (SGBR)

Paper 3: CapyMOA: Efficient Machine Learning for Data Streams in Python

Authors: Heitor M. Gomes et al. (2025)

CapyMOA is an efficient Python framework for machine learning on streaming data that integrates MOA and PyTorch into a unified continual learning environment. The library supports online classification, regression, anomaly detection, clustering, and concept drift detection while providing structured streaming pipelines, drift simulation, and prequential evaluation. It serves as a practical framework for developing real-time adaptive learning systems.

Requirement	Details
Research Question	How can an efficient framework support continual learning on streaming data?
Data Used	Synthetic and real streaming datasets (Hyper100K, CSV/ARFF streams)
Metrics Used	Accuracy, Runtime, Memory usage, Windowed evaluation
Key Findings	CapyMOA provides near-MOA efficiency with a flexible Python API for stream learning. 609804 PDF.pdf
Model	CapyMOA Online Learning Framework

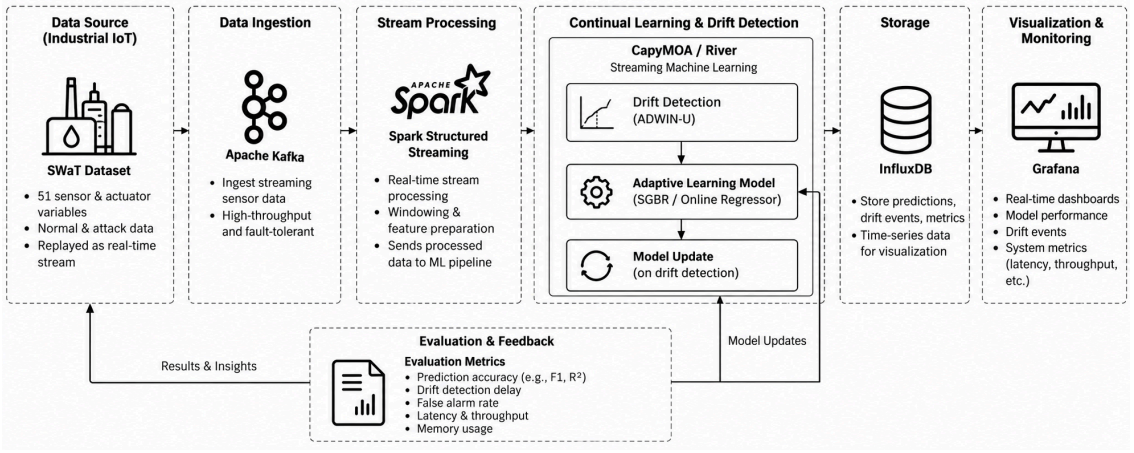
3. Dataset / Stream Source

SWaT (Secure Water Treatment Dataset) https://www.sutd.edu.sg/itrust/itrust-labs/datasets/dataset-characteristics/swat/

- Real industrial IoT sensor dataset
- 51 sensor and actuator variables
- Contains normal and anomalous operating conditions
- Stream replayed through Apache Kafka to simulate real-time data ingestion

4. Chosen Tool Combination

Pipeline Stage	Tool
Data Ingestion	Apache Kafka
Stream Processing	Spark Structured Streaming
Continual Learning	CapyMOA / River
Drift Detection	ADWIN-U
Time-Series Storage	InfluxDB
Visualization	Grafana



5. Evaluation Metrics

Metric	Purpose
Accuracy / F1 Score	Prediction quality
Drift Detection Delay	Speed of detecting concept drift
False Alarm Rate	Reliability of drift detector
Latency (ms)	Real-time processing performance
Throughput (events/sec)	Streaming capability
Memory Usage	Continual learning efficiency

6. Expected Enhancement

Baseline to Reproduce: ADWIN-based continual learning pipeline using CapyMOA for supervised drift detection.

Proposed Enhancement: Develop a Hybrid Continual Learning Framework that combines ADWIN-U (unsupervised drift detection) with Streaming Gradient Boosted Regression (SGBR) to enable adaptive model updates while minimizing the need for labeled data. The enhanced system will be compared against the baseline using accuracy, drift detection delay, latency, and memory usage.

References

Assis, D. N., & Souza, V. M. (2025). ADWIN-U: adaptive windowing for unsupervised drift detection on data streams: DN Assis, VMA Souza. *Knowledge and Information Systems*, *67*(11), 10005-10034.

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